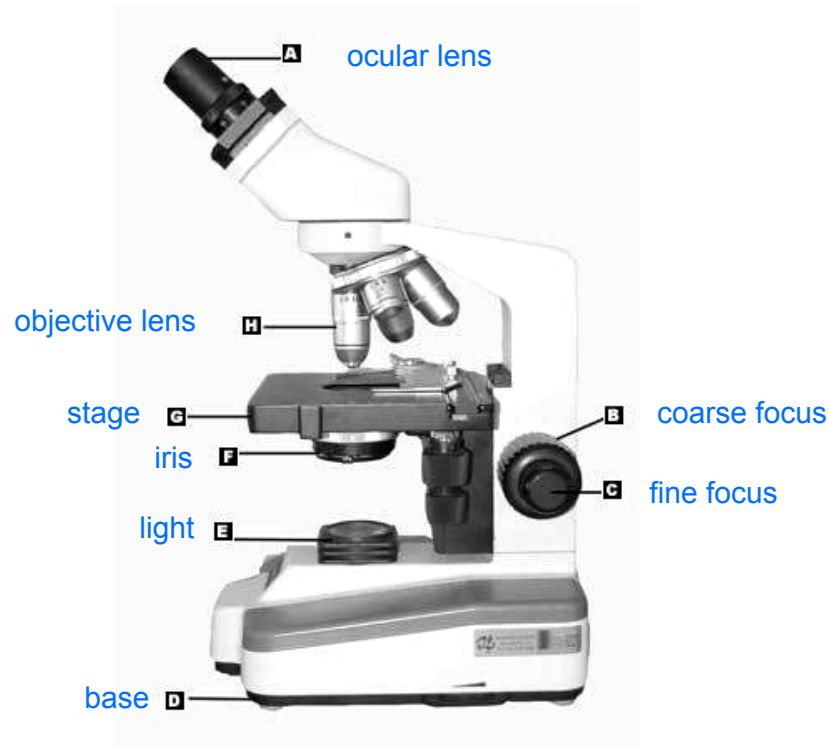


**YEAR 8 SCIENCE
BIOLOGICAL SCIENCE
REVISION - MICROSCOPES + CELLS**

1. (a) Label the following diagram of a microscope.



- (b) It is suggested that you look at the slide from the side when switching from low to high power. Why is this?

To make sure that the objective lens does not hit the slide. It will damage the lens.

- (c) If the eyepiece is X10 and the objective lens is X4, what is the magnification of the slide?

magnification = $10 \times 4 = X40$

- (d) If you look into the microscope and observe the slide in focus, what happens to the image if you move the slide to the **right**?

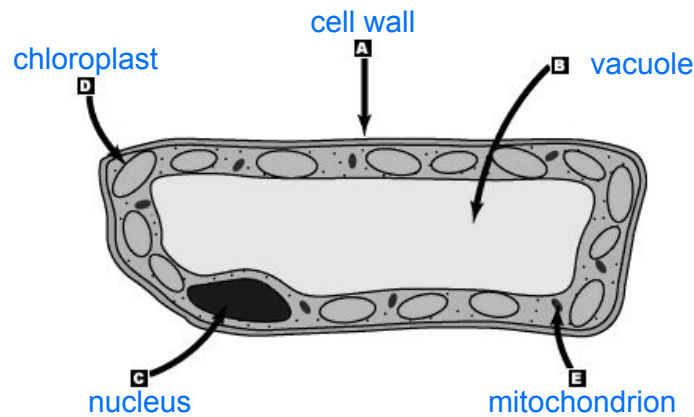
Moves left.

- (e) The microscopes used at Holy Cross have three objective lenses that can be adjusted to give low, medium and high power viewing of slides.

When is it best to use the **fine focus knob** when viewing a slide? Explain why.

When using high power. It moves the objective lens very small amounts and avoids crushing it into the slide, damaging it.

2. (a) Label the following diagram of a cell.



- (b) Is it a plant or animal cell? How do you know?

Plant cell - has a large vacuole, chloroplasts, cell wall.

- (c) Name the parts of the cell that do the following functions.

Part of Cell	Function
nucleus	Controls the activities of the cell.
vacuole	Storage area for substances needed by the cell.
cell membrane	Controls substances entering and leaving cells.
chloroplast	Contains chlorophyll and is the site for photosynthesis.
cell wall	Only found in plants – gives the cell strength and helps plants stay upright.
mitochondrion	Produces energy for the cell.
cytoplasm	Jelly-like liquid that fills up most of the space in a cell.

3. List three ways in which a plant cell is different to an animal cell.

- Have a cell wall.
- Contain a large vacuole.
- Have chloroplasts.

4. Unscramble the words using the clues provided.

- (a) Control centre of the cell. SEUNCLU nucleus
- (b) Surrounds the cell. ERAMMBNE membrane
- (c) Contains cell sap. OCVAUEL vacuole
- (d) Part of the cell between the cell membrane and the nucleus. CATOPLMYS cytoplasm
- (e) Building blocks of all living things. LELSC cells
- (f) Living things . ASMOGNIRS organisms

5. Copy and complete the table below.

Cell feature	Plant cells	Animal cells	Fungal cells
Cell wall	✓	✗	✓
Cytoplasm	✓	✓	✓
Cell membrane	✓	✓	✓
Chloroplast	✓	✗	✗
Nucleus	✓	✓	✓
Large vacuole	✓	✗	✗